
TRAINING SURGEONS WITH MACHINE LEARNING

Presented by:

Brendon Mustachio, Junior, Major in Computer Engineering

Selvin White, Sophomore, Major in Computer Science, IT intern

To applying machine learning to medical technology so surgeons can act fast and acquire useful information while they remain in control of a patient's care.

PROBLEM

Can ML models train surgeons efficiently and effectively?



- The application of machine learning in the medical field has made it easier for medical professionals to identify diseases and afflictions.
- ML helps to detect diseases earlier by examining X-ray and MRIs.
- Introduce an advanced model capable of processing live camera feed to reduce noise and sharpen contrast.
- Machine learning helps young surgeons perform better and increases their competence in an actual live surgery.
- Provide exact precision with surgical tools, reducing risk of accidental damage.

SOLUTION

Create a model capable of taking live motion data and providing timely feedback on technique

Example:

The Da Vinci surgery robot → → →

- Surgeon-controlled robotic platform
- Provides precise instrument movement
- Minimally invasive access

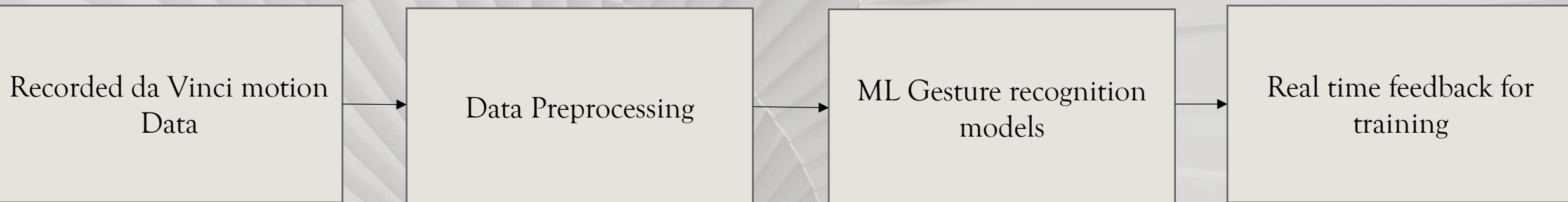


Our proposed machine learning system would support the surgeon by analyzing recorded robotic motion data and identifying surgical gestures during training tasks.

Tools:

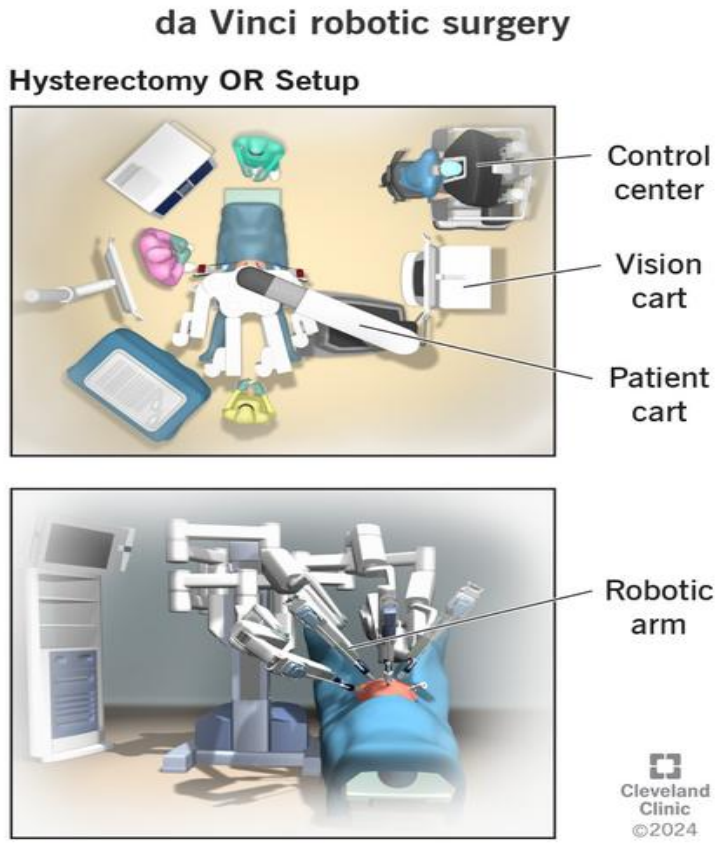
- Cameras
- Motion Simulator
- Python + OpenCV + TensorFlow/PyTorch

Proposed ML Workflow Plan



EVALUATION

- Use recorded Da Vinci surgical training motion data, including task such as suturing, knot tying, and needle passing.



- Compare a simple baseline model with temporal machine learning models.
- Measure gesture recognition accuracy and use a confusion matrix to identify commonly confused gestures.
- Measure the average latency, throughput, memory use, and model size to determine whether the system could support real time training feedback.
- Each configuration will run at least 3 times and will be reporting the averages and standard deviation.
- Expect a larger model to be more accurate while a smaller model may give a better balance of accuracy, response time, and memory usage.

FEEDBACK AND DISCUSSION

Sources:

<https://demigos.com/blog-post/machine-learning-in-the-medical-field-cases-challenges/>

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https://cirl.lcsr.jhu.edu/research/hmm/datasets/jigsaws_release/?utm_source=chatgpt.com

<https://www.uchealth.com/services/robotic-surgery/patient-information/davinci-surgical-system/>